Bhartiya Vidya Bhavan's

Sardar Patel Institute of Technology

Department of Computer Engineering

Academic Year 2026-27

Mini Project - Weekly Progress Report

Mini Project Title: Dynamic Timetable: A Role-Based Timetable Monitoring System with Automated Timetable Extraction and Natural-Language Queries

Name of Students:
1. Shivam Mulani (UID 2024300141)
2. Gurshaan Nandrajog (UID 2024300146)
3. Harshvarddhan Waghela (UID 2024300272)

Name of the Supervisor: Prof. Anant Nimkar

Review Date	Details of Work Done and Review of Meeting	Students Signature	Remarks by Supervisor	Supervisor Signature
17-08-2026	**Phase I: Problem identification and topic finalisation** • Studied the department's published 10-page timetable PDF and listed the problems students and faculty face: manual lookup, stacked lab batches, page-wise legends, clashes found late, old and revised versions circulating together. • Finalised the title, objectives and scope: read the published PDF, role-based views, plain-English questions, clash detection, free online demo. • Work split agreed: Shivam (architecture, database, security, query engine), Gurshaan (extraction, clash detection, performance), Harshvarddhan (mobile/web app, testing, documentation).			

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02-09-2026	**Phase II: Literature survey and requirements** • Surveyed 15 papers on university timetabling, table recognition/OCR, natural-language interfaces to databases, prompt/SQL injection, interval reasoning and RBAC; compared FET/UniTime/aSc, Tabula/Camelot and LLM text-to-SQL and identified the research gap. • Prepared the SRS: functional requirements for 5 roles (Student, Professor, HOD, Principal, Admin) and non-functional targets (at least 95% classes verified, answer in under 300 ms).			
30-09-2026	**Phase III: System design** • Designed the architecture (Expo app, FastAPI back end, background worker, PostgreSQL), DFDs, use-case, class, sequence and ER diagrams. • Designed the database schema (31 tables, 3NF), the REST API and the extraction and clash-detection algorithms; finalised the technology stack.			
12-10-2026	**Phase IV: Implementation and testing** • Shivam: schema and migrations, Argon2id login with rotating JWT tokens, role-based access, admin master data, natural-language query engine with clarification. • Gurshaan: extraction from PDF, OCR, Excel and Word; tuned on the real college PDF (lab batches, legends, noon-row label, combined B.E. page); time-row backtracking; clash detection (542 raw overlaps reduced to 2 genuine clashes). • Harshvarddhan: Expo app with role dashboards, weekly timetable, Ask screen, upload, review and admin screens. • Results on the real PDF: 452 of 453 classes verified (99.8%); automated back-end, app and end-to-end tests passing.			

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04-11-2026	**End Semester Evaluation (ESE)** • Since Phase IV: one student login per lab group, laptop browser support, Expo SDK 58, free online demo on Render, security fix for password reset. • Improvements after review: compositional query planner (17 operations combined with filters, dates and time windows), teachers found by code, first name, surname or 'X sir', subject typos matched, anyone can view any teacher's, batch's or room's timetable (personal timetables stay private). • Measured: 37-47 ms median answer time, 0 wrong answers with 32 parallel clients; 179 back-end + 20 app tests and 45 end-to-end checks passing. • Deliverables: report on the department LaTeX template, A3 poster, IEEE-format research paper, presentation, scenario-based challenges; live demo.			